

Path Signatures for Regime Detection in Cryptocurrency Markets: A Rough-Path Framework Using Spot, Perpetual Basis, and Funding Rates

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April 19, 2026

Abstract

We study market regime detection in cryptocurrency markets through the lens of rough path theory. Classical approaches — hidden Markov models (HMMs) on returns, Gaussian mixture models, and rolling-window volatility thresholds — impose Markovian or memoryless dynamics and summarize path information through low-dimensional moment statistics. In cryptocurrency markets, however, regime transitions are typically driven by reflexive leverage cycles, coordinated liquidations, and perpetual-swap funding imbalances: path-dependent, non-Markovian, multivariate phenomena that these approaches systematically underfit. We propose an alternative based on *truncated path signatures*, iterated-integral features of transformed price paths originating in the rough-path theory of Lyons [1998]. Our contributions are fourfold. First, we construct a crypto-native multivariate path combining BTC and ETH log-prices, a realized-volatility proxy, the perpetual-spot basis, and hourly funding rates, augmented with lead-lag and time transformations; basis and funding are directly observable measures of leverage positioning and, to our knowledge, have not previously appeared as signature coordinates. Second, we prove a universal-approximation bound (Theorem 3.4) for continuous regime-labeling functionals by linear functionals of truncated signatures, with an explicit depth bound in terms of the modulus of continuity of the target. Third, we benchmark signature-based regime features against Gaussian HMMs and rolling-volatility detectors on strictly out-of-sample downstream tasks using hourly Binance data, January 2020 through April 2026, approximately 5.5×10^4 observations. Fourth, we perform an event study across six canonical regime shifts (COVID-19, May 2021, Terra-LUNA, FTX, SVB/USDC, BTC ETF approval). Signature features deliver a 5–7 percentage-point accuracy advantage over returns-based HMM baselines on forward-volatility regime classification; the advantage localizes in third-order iterated integrals, validating the theoretical prediction; and signatures correctly anticipate the out-of-sample SVB/USDC event and correctly abstain on the bullish BTC ETF approval. Signatures do not solve directional return or drawdown prediction at sub-daily horizons, consistent with market-efficiency expectations. Code, data pipelines, and trained-model artifacts are released as a reproducible reference implementation.

Keywords: rough path theory, path signatures, regime detection, cryptocurrency, perpetual futures, funding rates, hidden Markov models, universal approximation.

JEL Classification: C14, C38, C58, G17. **MSC 2020:** 60L10, 60L20, 62M10, 91G70.

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1. Introduction

1.1. Motivation

Cryptocurrency markets have, over the past decade, matured into a liquid, globally accessible, continuously-traded asset class. Daily spot turnover on the largest centralized exchanges regularly exceeds fifty billion U.S. dollars, and notional open interest in perpetual futures on Bitcoin and Ether often surpasses the corresponding cash-settled equity-index contracts traded on CME. Despite this scale, the statistical behavior of crypto returns remains markedly distinct from traditional financial assets. Unconditional volatility is three to five times that of large-capitalization equities; return distributions exhibit heavier tails and sharper peaks; trading runs twenty-four hours a day with no weekend or overnight closures; and the market microstructure features deep perpetual-swap order books in which leverage is directly observable through exchange-reported funding rates and open-interest time series.

These differences are not merely quantitative. They reflect a qualitative change in the mechanisms that drive price dynamics. Whereas equity regime shifts typically propagate through earnings cycles, monetary-policy announcements, or slow-moving macro narratives, crypto regime shifts are frequently *reflexive* in the sense of [Soros \[1987\]](#): a change in aggregate leverage positioning precipitates a forced-liquidation cascade, which alters the observed price path, which in turn feeds back into positioning. The collapses of Terra-LUNA in May 2022 and of FTX in November 2022 are the canonical examples. In both episodes, the transition from a stable or modestly-trending regime into acute distress occurred on timescales of hours rather than weeks, and the statistical signatures of the coming transition were visible in the *joint* dynamics of spot prices, perpetual basis, funding rates, and cross-asset correlations well before they registered in any univariate volatility summary.

Detecting such regime shifts *ex ante*, or with minimal *ex-post* lag, is consequential for several classes of market participant. Exchange risk desks, prime brokers, and centralized market-makers require real-time regime signals to tighten margin and inventory limits before liquidation cascades become self-reinforcing. Quantitative portfolio managers benefit from regime-conditional allocation rules, since the optimal tilt between mean-reverting and trend-following strategies depends strongly on the prevailing regime [[Ang and Bekaert, 2002](#), [Guidolin and Timmermann, 2007](#)]. Execution algorithms require regime awareness to adapt participation rates and slippage expectations. In each of these applications, the cost of a missed or delayed detection is measurable in basis points of P&L or in dollars of avoidable liquidation; the value of faster or more accurate detection is correspondingly high.

1.2. Inadequacy of classical methods in the crypto setting

The standard toolkit for regime detection in financial time series was developed decades before the existence of crypto markets and rests on assumptions that are violated in this setting. Three families of methods dominate practice. Hidden Markov models, introduced in their economic form by [Hamilton \[1989\]](#), posit an unobservable discrete state evolving as a Markov chain and conditioning the distribution of observed returns; their inference machinery — Baum-Welch estimation [[Baum et al., 1970](#)] and Viterbi decoding [[Viterbi, 1967](#)] — is mature and widely implemented. Gaussian mixture models and their Markov-switching extensions [[Frühwirth-Schnatter, 2006](#)] relax the transition structure but retain conditional independence of observations given the state. Rolling-window statistics on realized volatility, drawdown, or price-to-moving-average ratios [[Pagan and Sossounov, 2003](#), [Lunde and Timmermann, 2004](#)] provide nonparametric threshold detectors that are standard on trading desks.

All three families share two features that are limiting in crypto. *First*, they are Markovian or memoryless with respect to path shape: the state at time t is conditionally independent of the observed past given the state at $t - 1$ in the HMM, or given the current observation in a threshold rule. This explicitly excludes the path-dependent, cumulative-leverage-driven dynamics that dominate crypto regime transitions. A leverage cycle builds over days or weeks and manifests as a characteristic geometric shape in the joint trajectory of basis, funding, and open interest; a Markov model conditioned only on recent returns cannot, by construction, discriminate such a buildup from an otherwise-similar distribution of returns. *Second*, these methods summarize multivariate information through low-dimensional moment statistics, typically mean and variance of returns over a window. This discards the *order-aware interaction structure* between coordinates. A period in which BTC spot and ETH basis widen *together*, driven by long-side leverage build-up, is statistically distinguishable from a period in which they widen *in opposing sequence*, driven by short-side hedging flows — yet a second-moment summary cannot make this distinction.

1.3. Rough paths and signatures

A mathematically principled response to the limitations above is provided by the theory of *rough paths* [Lyons, 1998, Lyons et al., 2007, Friz and Victoir, 2010]. Given a continuous path $X : [0, T] \rightarrow \mathbb{R}^d$ of finite p -variation, the *signature* of X is the infinite sequence of iterated integrals

$$S(X)_{0,T} = \left(1, \int_{0 < t_1 < T} dX_{t_1}, \iint_{0 < t_1 < t_2 < T} dX_{t_1} \otimes dX_{t_2}, \iiint dX_{t_1} \otimes dX_{t_2} \otimes dX_{t_3}, \dots\right),$$

valued in the tensor algebra $T((\mathbb{R}^d)) = \bigoplus_{k=0}^{\infty} (\mathbb{R}^d)^{\otimes k}$. The construction dates to Chen [1957] for smooth paths and was extended by Lyons to paths of arbitrary finite p -variation. Four properties of the signature jointly motivate its use as a feature map for time-series learning: faithfulness (injectivity up to tree-like equivalence after time augmentation; Hambly and Lyons, 2010); universal approximation [Levin et al., 2013, Chevyrev and Oberhauser, 2022, Fermanian, 2022]; reparameterization invariance of the untruncated signature; and tractable finite-dimensional truncation. Efficient implementations — `iisignature` [Reizenstein and Graham, 2020] and `signatory` [Kidger and Lyons, 2020] — make signature feature extraction routine on commodity hardware.

These properties have motivated a rapidly growing literature applying signature methods in quantitative finance, including volatility modeling [Alòs et al., 2025, Bayer et al., 2016], option pricing and hedging [Kalsi et al., 2020, Cuchiero et al., 2023], volatility forecasting, and time-series generation [Buehler et al., 2020]. Most closely related to the present work, Horvath and Issa [2023] develop a non-parametric online regime-detection method using signatures as a feature map within a maximum-mean-discrepancy two-sample test.

1.4. Gap and contributions

Despite the expanding literature on signatures in finance, applications to cryptocurrency markets remain sparse. Recent work applies signatures to cluster digital assets for portfolio construction — a cross-sectional similarity problem — but does not address *temporal* regime detection or exploit perpetual-futures microstructure. No rigorous published treatment of signature-based regime detection in crypto, using a crypto-native path construction with observable leverage signals and out-of-sample predictive validation, currently exists. This paper closes that gap.

Contribution 1 — Crypto-native path construction. We construct a multivariate path

$$X_t = (t, \log P_t^{\text{BTC}}, \log P_t^{\text{ETH}}, \text{RV}_t, b_t^{\text{BTC}}, b_t^{\text{ETH}}, f_t^{\text{BTC}}, f_t^{\text{ETH}}),$$

comprising time, log-spot prices, a high-frequency realized-volatility proxy, perpetual-spot basis, and hourly-annualized funding rate for each asset. The inclusion of basis and funding as signature coordinates — directly observable measures of leverage positioning not available in equity markets — is, to our knowledge, novel in the signature-methods literature. We additionally apply lead-lag embedding [Flint et al., 2016] and time augmentation to ensure cross-coordinate sequential dependence is captured and signature injectivity is guaranteed.

Contribution 2 — Theoretical guarantee. In Section 3 we prove (Theorem 3.4) that for any continuous regime-labeling functional $L : \mathcal{K} \rightarrow \mathbb{R}$ on a compact set of bounded-variation paths and any $\varepsilon > 0$, there exists a truncation depth N and a linear functional ℓ on $T^N((\mathbb{R}^d))^*$ such that $\sup_{X \in \mathcal{K}} |L(X) - \langle \ell, S^{\leq N}(X)_{0,T} \rangle| < \varepsilon$, with an explicit bound on N in terms of the modulus of continuity of L and the 1-variation diameter of $\mathcal{K}$. The proof combines the Stone-Weierstrass theorem, the shuffle-algebra structure of signatures, and the factorial decay of signature norms from Lyons’s extension theorem.

Contribution 3 — Predictive out-of-sample evaluation. We benchmark all methods on strictly out-of-sample downstream tasks: (A) forward realized-volatility tercile classification at horizons 24h, 72h, 168h; (B) forward 24-hour maximum-drawdown regression; (C) directional return classification at horizons 1h, 6h, 24h. All methods are fit on a pre-2023 training sample and evaluated on a strictly post-2023 test sample.

Contribution 4 — Event study. We quantify detection timing for each method across six canonical crypto regime shifts: COVID-19 (March 2020), the May 2021 leverage unwind, Terra-LUNA (May 2022), FTX insolvency (November 2022), SVB/USDC depeg (March 2023), and the U.S. spot Bitcoin ETF approval (January 2024). The event study provides an interpretable, method-agnostic view of practical detection performance that complements the aggregate out-of-sample statistics.

1.5. Organization

Section 2 surveys related work. Section 3 develops the mathematical background and states and proves our main theoretical result. Section 4 presents the methodology. Section 5 describes the data. Section 6 reports empirical results. Sections 7–8 discuss and conclude.

2. Related Work

2.1. Path signatures in quantitative finance

The theory of rough paths was introduced by Lyons [1998] to give a robust, pathwise meaning to controlled differential equations driven by signals of low regularity, generalizing the Itô-Stratonovich framework to deterministic and semi-deterministic settings. The central object, the signature, was studied earlier in the smooth-path setting by Chen [1957], who established its algebraic structure as an element of the tensor algebra and proved the shuffle identity linking products of iterated integrals to linear combinations of higher-order iterated integrals. The monographs of Lyons et al. [2007] and Friz and Victoir [2010] provide the standard references, including the extension theorem on factorial decay and the uniqueness-of-signature result of Hambly and Lyons [2010].

The translation of signature theory into a machine-learning toolkit was developed by Levin et al. [2013] for streaming prediction problems, formalized by Chevyrev and Kormilitzin [2016] in the feature-map perspective, and extended to probability measures on path space by Chevyrev and Oberhauser [2022] through the notion of signature moments. Fermanian [2022] provides a comprehensive treatment of signature-based learning with statistical consistency results.

Financial applications have proliferated. Gyurkó et al. [2014] applied signature features to high-frequency equity data and showed predictive gains over standard technical-indicator baselines. Kalsi et al. [2020] studied signature methods for execution under rough-path dynamics; Cuchiero et al. [2023] extended this to a class of signature-based stochastic models. Buehler et al. [2020] introduced a data-driven market simulator in which generated sample paths match empirical signature moments. Alòs et al. [2025] compare signature-based and Malliavin-expansion approaches to implied volatility calibration. The most directly relevant prior work is Horvath and Issa [2023], who propose a non-parametric online regime-detection procedure based on the signature maximum-mean-discrepancy; their illustrations are on equity and FX data, not crypto. We adopt the signature-MMD framework as one signature-based detection variant in our methodology.

2.2. Regime detection in financial time series

The formal modeling of regime changes in financial time series has a long history. Hamilton [1989, 1990] introduced Markov-switching autoregressive models. Kim [1994] unified Markov-switching with state-space dynamics; Frühwirth-Schnatter [2006] provides the Bayesian treatment. Ang and Bekaert [2002] documented regime-dependent dynamics in global interest rates; Guidolin and Timmermann [2007] applied Markov-switching models to international asset allocation. Ang and Timmermann [2012] survey the broader literature.

A second tradition defines regimes through ex-post statistical criteria. Pagan and Sossounov [2003] formalized the Bry-Boschan algorithm for dating bull and bear markets; Lunde and Timmermann [2004] studied duration dependence in such phases. Threshold-based indicators remain ubiquitous on trading desks. The changepoint-detection literature [Basseville and Nikiforov, 1993, Truong et al., 2020] provides online distributional-break detection; the MMD test of Gretton et al. [2012] extends this to non-parametric distributional change.

2.3. Cryptocurrency market modeling

Empirical study of cryptocurrency return dynamics has matured. Urquhart [2016] documented significant departures from weak-form efficiency in Bitcoin returns. Katsiampa [2017] surveyed GARCH-family specifications and identified long-memory and leverage-asymmetry patterns. Makarov and Schoar [2020] documented persistent cross-exchange arbitrage spreads and analyzed their connection to capital-control frictions. Griffin and Shams [2020] examined the influence of Tether on Bitcoin pricing during the 2017 bull market.

A rough-volatility perspective on crypto has emerged more recently. Takaishi [2020] and Bianchi et al. [2022] estimate Hurst exponents of log-volatility for Bitcoin and find values substantially below $H = 1/2$, consistent with the rough-volatility paradigm of Bayer et al. [2016] and Gatheral et al. [2018]. These studies establish that crypto log-volatility is substantially rougher than equity benchmarks, providing motivation for rough-path methods that accommodate low-regularity dynamics natively. Regime-switching models applied to crypto [Ardia et al., 2019, Caporale and Zekokh, 2019] are methodologically conventional — returns-based HMMs and Markov-switching GARCH — and do not leverage multivariate perpetuals data or path-dependent feature representations.

2.4. Positioning

We close three gaps. First, the signature-methods literature in finance has matured, but applications to crypto remain isolated to a single cross-sectional clustering study that uses neither perpetuals microstructure nor temporal regime-detection framing. Second, the classical regime-detection literature in crypto is confined to returns-based Markov-switching models that cannot exploit the

joint-path information in spot, basis, and funding. Third, the rough-volatility literature for crypto establishes that low-regularity pathwise methods are theoretically appropriate but has not been combined with signature-based feature representations for downstream prediction. We construct a crypto-native multivariate path with lead-lag and time augmentation, prove a quantitative universal-approximation bound, and evaluate predictively on strictly out-of-sample data.

3. Mathematical Background

3.1. Paths and p -variation

Fix a horizon $T > 0$ and dimension $d \geq 1$. A *path* is a continuous map $X : [0, T] \rightarrow \mathbb{R}^d$. For a partition $\pi = \{0 = t_0 < \dots < t_n = T\}$ and $p \geq 1$, the p -variation of X on $[s, t] \subseteq [0, T]$ is

$$\|X\|_{p\text{-var}, [s, t]} = \left(\sup_{\pi} \sum_i \|X_{t_{i+1}} - X_{t_i}\|^p \right)^{1/p}.$$

Denote by $BV_p([0, T]; \mathbb{R}^d)$ the Banach space of continuous paths of finite p -variation. Financial time series are observed at discrete times; we convert them to continuous paths by linear interpolation, so BV_1 is the natural ambient space.

3.2. The tensor algebra and the signature

The tensor algebra over $\mathbb{R}^d$ is $T((\mathbb{R}^d)) = \bigoplus_{k \geq 0} (\mathbb{R}^d)^{\otimes k}$. Fix an orthonormal basis $\{e_1, \dots, e_d\}$ of $\mathbb{R}^d$; for a multi-index $I = (i_1, \dots, i_k)$, write $e_I = e_{i_1} \otimes \dots \otimes e_{i_k}$. The truncated tensor algebra at depth N is

$$T^N((\mathbb{R}^d)) = \bigoplus_{k=0}^N (\mathbb{R}^d)^{\otimes k}, \quad \dim T^N = \sum_{k=0}^N d^k = \frac{d^{N+1} - 1}{d - 1}.$$

For $X \in BV_1([0, T]; \mathbb{R}^d)$, the *signature* of X on $[s, t]$ is

$$S(X)_{s,t} = (1, S_{s,t}^1, S_{s,t}^2, \dots), \quad S_{s,t}^k(X) = \iint_{s < t_1 < \dots < t_k < t} \dots \int dX_{t_1} \otimes \dots \otimes dX_{t_k}.$$

Coordinate-wise, for a multi-index $I = (i_1, \dots, i_k)$,

$$\langle S(X)_{s,t}, e_I \rangle = \iint_{s < t_1 < \dots < t_k < t} \dots \int dX_{t_1}^{i_1} dX_{t_2}^{i_2} \dots dX_{t_k}^{i_k}. \quad (1)$$

Signature coordinates encode an infinite collection of *order-aware* interactions. The $(1, 2)$ -coordinate $\iint_{t_1 < t_2} dX_{t_1}^1 dX_{t_2}^2$ differs in general from $(2, 1)$; their difference equals twice the signed Lévy area enclosed by the two-dimensional projection of X onto the (X^1, X^2) -plane. Such coordinates distinguish paths differing only in the *order* of coordinate movements — a distinction moment-based summaries cannot make.

Proposition 3.1 (Factorial decay). *For $X \in BV_1([0, T]; \mathbb{R}^d)$ and all $k \geq 0$,*

$$\|S_{s,t}^k(X)\| \leq \frac{\|X\|_{1\text{-var}, [s, t]}^k}{k!},$$

where $\|\cdot\|$ is the Hilbert-Schmidt norm on $(\mathbb{R}^d)^{\otimes k}$. Consequently

$$\|S(X)_{s,t} - \pi^N S(X)_{s,t}\| \leq \sum_{k > N} \|X\|_{1\text{-var}}^k / k!.$$

Proof. Let $\omega(s, t) = \|X\|_{1\text{-var}, [s, t]}$. By Fubini and the definition of the iterated integral,

$$\|S_{s, t}^k(X)\| \leq \iint_{s < t_1 < \dots < t_k < t} \dots \int \|dX_{t_1}\| \dots \|dX_{t_k}\| = \frac{1}{k!} \left(\int_s^t \|dX_u\| \right)^k = \frac{\omega(s, t)^k}{k!},$$

with the $1/k!$ factor arising from the volume of the ordered simplex. $\square$

For typical post-normalization crypto windows with per-window 1-variation $M \in [0.3, 1.0]$, truncation at $N = 4$ retains at least $1 - \sum_{k \geq 5} 1^k/k! \approx 99.3\%$ of the signature norm, recovering standard practical truncation guidance [Fermanian, 2022] as a quantitative consequence of path regularity.

3.3. Chen's identity and the shuffle relation

Proposition 3.2 (Chen, 1957). *For $0 \leq s \leq u \leq t \leq T$,*

$$S(X)_{s, t} = S(X)_{s, u} \otimes S(X)_{u, t}. \quad (2)$$

Chen's identity makes the signature a multiplicative functional on the simplex of time intervals and yields a practical computation algorithm for piecewise-linear paths.

Proposition 3.3 (Shuffle identity; Ree, 1958). *For multi-indices $I = (i_1, \dots, i_m)$ and $J = (j_1, \dots, j_n)$, let $I \cdot J$ denote the set of $(m + n)$ -element multi-indices obtained by interleaving I and J while preserving each index's internal order. Then*

$$\langle S(X), e_I \rangle \cdot \langle S(X), e_J \rangle = \sum_{K \in I \cdot J} \langle S(X), e_K \rangle. \quad (3)$$

The shuffle identity implies that linear functionals of the signature form a polynomial algebra: linear regression on signatures realizes polynomial regression on path coordinates.

3.4. Path augmentations

Time augmentation. The time-augmented path $\tilde{X}_t = (t, X_t^1, \dots, X_t^d)$ on $\mathbb{R}^{d+1}$ has strictly monotone first coordinate; by Hambly and Lyons [2010], this excludes tree-like equivalence and renders $X \mapsto S(\tilde{X})$ injective on piecewise-linear paths.

Lead-lag transformation. For X sampled at $0 = t_0 < \dots < t_n = T$, the lead-lag transform $\hat{X} : [0, 2n] \rightarrow \mathbb{R}^{2d}$ is the piecewise-linear path through $\hat{X}_{2i} = (X_{t_i}, X_{t_i})$ and $\hat{X}_{2i+1} = (X_{t_{i+1}}, X_{t_i})$. Level-two cross-coordinate terms of the lead-lag signature recover the quadratic covariation of X [Flint et al., 2016], and more generally encode the sequential structure of cross-coordinate movements.

3.5. Signature kernels and MMD

For paths $X, Y \in BV_1$, the *signature kernel* is $k_{\text{sig}}(X, Y) = \langle S(X)_{0, T}, S(Y)_{0, T} \rangle$. Given probability measures μ, ν on path space, the signature MMD is

$$\text{MMD}_{\text{sig}}(\mu, \nu) = \left\| \mathbb{E}_{X \sim \mu}[S(X)] - \mathbb{E}_{Y \sim \nu}[S(Y)] \right\|.$$

Chevyrev and Oberhauser [2022] showed that under normalization, MMD_{sig} is characteristic: $\text{MMD}_{\text{sig}}(\mu, \nu) = 0$ iff $\mu = \nu$. This underlies the Horvath-Issa regime-detection procedure.

3.6. Main theorem

Theorem 3.4 (Uniform approximation of regime functionals). *Let $\mathcal{K} \subset BV_1([0, T]; \mathbb{R}^d)$ be a compact set of continuous time-augmented paths. Let $L : \mathcal{K} \rightarrow \mathbb{R}$ be continuous in the 1-variation metric, and let $M = \sup_{X \in \mathcal{K}} \|X\|_{1-var, [0, T]}$. Then for every $\varepsilon > 0$ there exist a truncation depth $N \in \mathbb{N}$ and a linear functional $\ell \in (T^N((\mathbb{R}^d)))^*$ such that*

$$\sup_{X \in \mathcal{K}} |L(X) - \langle \ell, \pi^N S(X)_{0,T} \rangle| < \varepsilon. \quad (4)$$

The depth N may be chosen to satisfy $\sum_{k>N} M^k/k! < \varepsilon/(2\|\ell\|_*)$, where $\|\ell\|_*$ is the dual norm of ℓ .

Proof. Define $S : \mathcal{K} \rightarrow T((\mathbb{R}^d))$ by $X \mapsto S(X)_{0,T}$ and let $\tilde{\mathcal{K}} = S(\mathcal{K})$.

Step 1 (Injectivity). By Hambly and Lyons [2010], identical signatures imply tree-like equivalence. Time augmentation excludes tree-like equivalence, so S is injective on $\mathcal{K}$.

Step 2 (Compactness). By Proposition 3.1, S is continuous from $\mathcal{K}$ (in 1-variation topology) into $T((\mathbb{R}^d))$ (in norm). The continuous image of a compact set is compact, so $\tilde{\mathcal{K}}$ is compact.

Step 3 (Dense subalgebra). Let $\mathcal{A} \subset C(\tilde{\mathcal{K}}; \mathbb{R})$ be the subalgebra generated by the constants and the coordinate evaluations $s \mapsto \langle e_I, s \rangle$ for all finite multi-indices I . By the shuffle identity (3), products of coordinate evaluations are linear combinations of coordinate evaluations at higher indices, so $\mathcal{A}$ coincides with the linear functionals on the full signature. $\mathcal{A}$ contains the constants and separates points of $\tilde{\mathcal{K}}$ (by Step 1, distinct elements of $\mathcal{K}$ have distinct signatures; coordinate functionals jointly separate points). By the Stone-Weierstrass theorem [Rudin, 1976, Thm. 7.32], $\mathcal{A}$ is uniformly dense in $C(\tilde{\mathcal{K}}; \mathbb{R})$.

Step 4 (Depth bound). Since $L : \mathcal{K} \rightarrow \mathbb{R}$ is continuous and $S|_{\mathcal{K}}$ is a homeomorphism onto $\tilde{\mathcal{K}}$, $\tilde{L} = L \circ S^{-1} : \tilde{\mathcal{K}} \rightarrow \mathbb{R}$ is continuous. By Step 3, for every $\varepsilon > 0$ there exists a finite linear combination $\ell \in T((\mathbb{R}^d))^*$ involving only finitely many coordinates such that

$$\sup_{s \in \tilde{\mathcal{K}}} |\tilde{L}(s) - \langle \ell, s \rangle| < \varepsilon/2. \quad (5)$$

Let N be the maximum level of any coordinate appearing in ℓ . Then $\ell \in (T^N((\mathbb{R}^d)))^*$. By the triangle inequality and Proposition 3.1,

$$|\langle \ell, S(X) \rangle - \langle \ell, \pi^N S(X) \rangle| \leq \|\ell\|_* \cdot \sum_{k>N} M^k/k!.$$

Choosing N large enough that $\sum_{k>N} M^k/k! < \varepsilon/(2\|\ell\|_*)$ and combining with (5) yields (4). $\square$

Remark 3.5 (Practical truncation). For our hourly crypto path with per-window $M \in [0.3, 1.0]$ (Section 5), depth $N = 3$ bounds truncation error at $\leq M^4/4! \leq 0.042$, and $N = 4$ at ≤ 0.0083 . The dimension of the truncated-signature vector for $d_{\text{aug}} = 15$ at these depths is 3,615 and 54,240, respectively. We use $N = 3$ as the main specification for computational tractability, with truncation error well below the typical magnitude of signal we detect.

The practical implication of Theorem 3.4 is that extracting a truncated signature and fitting a linear classifier is *universally approximating* for any continuous regime-labeling target.

4. Methodology

4.1. Pipeline overview

Let $\mathcal{D} = \{\mathbf{r}_t\}_{t=1}^{T_{\max}}$ denote the multivariate hourly panel described in Section 5, with $\mathbf{r}_t \in \mathbb{R}^{d_{\text{raw}}}$ and $d_{\text{raw}} = 7$: BTC and ETH hourly log-returns, a 24-hour realized-volatility proxy, BTC and ETH perpetual-spot basis, and BTC and ETH 8-hour funding rates. The pipeline proceeds as follows.

1. *Windowing.* Rolling windows $W_t = \{\mathbf{r}_{t-W+1}, \dots, \mathbf{r}_t\}$ with $W = 168$ (one week), stride $\Delta = 1$ (hourly).
2. *Path construction.* For each window, integrate return coordinates, pass level coordinates through, basepoint-translate, lead-lag embed ($2d_{\text{raw}} = 14$ coordinates), and prepend time ($d_{\text{aug}} = 15$).
3. *Signature extraction.* Compute truncated signature $s^{(t)} = \pi^N S(\tilde{X}^{(t)})_{0,W} \in \mathbb{R}^{D_N}$, $D_3 = 3,615$.
4. *Preprocessing.* Level-wise factorial normalization; robust-scaling using training-fold IQRs; truncated SVD to 128 components (explains 99.3% of training-fold variance).
5. *Modeling.* Fit signature-based or baseline model (Section 4.3) with targets from one of three downstream tasks (Section 4.4).
6. *Evaluation.* Strictly out-of-sample prediction on 2023–2026 test sample with metric suite of Section 4.4.

4.2. Path construction and augmentations

For a window $\{\mathbf{r}_{t-W+1}, \dots, \mathbf{r}_t\}$ we construct the pre-augmentation path by integrating return coordinates and passing level coordinates through:

$$Y_k^{(t)} = \left(\sum_{j=1}^k r_{t-W+j}^{\text{BTC}}, \sum_{j=1}^k r_{t-W+j}^{\text{ETH}}, v_{t-W+k}, b_{t-W+k}^{\text{BTC}}, b_{t-W+k}^{\text{ETH}}, f_{t-W+k}^{\text{BTC}}, f_{t-W+k}^{\text{ETH}} \right).$$

Basis and funding are levels, not increments: integrating them would destroy the path-relevant information in their levels. We linearly interpolate, basepoint-translate to $Y_0^{(t)} = \mathbf{0}$, apply lead-lag to $\mathbb{R}^{14}$, and prepend a time coordinate.

Scale normalization. Raw coordinates span scales from $O(10^{-4})$ (funding) to $O(10^{-2})$ (realized vol). We robust-scale each raw coordinate to unit IQR using training-fold statistics only. After signature extraction, level- k blocks are rescaled by $k!$ to undo factorial decay per Fermanian [2022, §5.2].

4.3. Models

We compare five methods, each attaching the same downstream learner (logistic regression or ridge) to a different feature representation.

- **SIG-LR.** 128-dimensional truncated-SVD representation of depth-3 signature features.
- **HMM-LR.** Filtered posterior $\gamma_t = (P(Z_t = k | \mathbf{r}_{1:t}^{\text{BTC}}))_k$ of a Gaussian HMM with $K = 3$ states fit by Baum-Welch on BTC hourly returns. Forward-filtered, not smoothed, to avoid lookahead.
- **HMM-MULTI-LR.** Filtered posterior of a Gaussian HMM with $K = 3$ diagonal-covariance states fit on the full 7-feature panel. The direct counterpart to SIG-LR with the same feature set but a Markovian representation.
- **ROLLVOL-LR.** One-hot tercile indicator based on trailing 168-hour realized volatility of BTC with tercile boundaries estimated on the training fold.
- **SIG+HMM-LR.** Concatenation of SIG and HMM-MULTI features.

4.4. Tasks and metrics

Task A: realized-volatility tercile classification. Target: tercile of BTC realized volatility over the next $h \in \{24, 72, 168\}$ hours. Metrics: accuracy, macro- F_1 , macro one-vs-rest ROC-AUC, log-loss. Terciles are defined using training-fold boundaries to prevent lookahead.

Task B: forward-drawdown regression. Target: maximum relative drawdown of BTC spot over the next 24 hours. Metrics: out-of-sample R^2 , RMSE, MAE, and tail R^2 computed on the worst-quartile realized drawdowns.

Task C: directional return classification. Target: sign of BTC return over the next $h \in \{1, 6, 24\}$ hours. Metrics: accuracy, AUC, Brier score, macro- F_1 .

4.5. Train-test protocol

Models are trained on the 26,081 windows with terminal hour in 2020-01-01 through 2022-12-31 and tested on the 28,870 windows with terminal hour in 2023-01-01 through 2026-04-17. All feature-scaling, SVD, tercile boundaries, HMM parameters, and regularization choices are fit on the training fold only. Signature features for window t use only $\{\mathbf{r}_s : s \leq t\}$; targets use only $\{\mathbf{r}_s : s > t\}$. No lookahead is possible by construction.

The implementation is Python 3.12 with `iisignature` 0.24, `hmmlearn` 0.3, `scikit-learn` 1.8. Signature extraction of 55,129 windows completes in 3.6 minutes on an 8-core machine.

5. Data

5.1. Sources and construction

All data are from Binance via `data.binance.vision`, which provides monthly and daily ZIP-compressed CSV archives of kline bars and per-contract funding-rate histories. We collect one-hour klines for BTC/USDT and ETH/USDT spot, one-hour klines for the corresponding USDT-margined perpetual contracts, and the full 8-hour funding-rate history for both perpetuals, covering January 1, 2020 through April 17, 2026. Timestamps are normalized to UTC; a subset of 2025 files use microsecond precision and are rescaled to millisecond-since-epoch.

The panel features are: BTC and ETH hourly log-returns $r_t^i = \log(P_t^i/P_{t-1}^i)$; a 24-hour trailing realized-volatility proxy $v_t = \sqrt{\sum_{j=0}^{23} (r_{t-j}^{\text{BTC}})^2} \cdot \sqrt{365}$; perpetual-spot basis $b_t^i = (F_t^i - P_t^i)/P_t^i$; and the 8-hour funding rate f_t^i forward-filled to the hourly grid. After dropping rows with initializing volatility, the usable sample is 55,129 hourly observations at 100% coverage.

5.2. Summary statistics

Table 1 reports marginal moments. BTC hourly log-returns have excess kurtosis 53.5 and ETH 28.6, versus 0 for Gaussian. The BTC perpetual-spot basis has excess kurtosis 161.7, driven by liquidation-cascade hours where the perpetual traded at a 200-300 bp discount to spot — an arbitrage-dislocation signal invisible in spot-return distributions. Funding rates are mean-positive and right-skewed, consistent with persistent long-side leverage demand.

5.3. Figures

Figure 1 displays BTC and ETH spot prices on a log vertical axis with the six canonical regime-shift events marked as vertical bands. Figure 2 plots perpetual-spot basis and hourly-annualized funding rates; the 2021 bull run is clearly visible as persistent strongly-positive funding, while the post-FTX

Table 1: Summary statistics, 55,129 hourly observations, 2020-01-01 through 2026-04-17.

Feature	Mean	Std	Median	95%	Skew	Excess kurt.
BTC log-return (hourly)	4.2×10^{-5}	6.73×10^{-3}	6.1×10^{-5}	9.19×10^{-3}	-0.94	53.5
ETH log-return (hourly)	5.2×10^{-5}	8.56×10^{-3}	8.2×10^{-5}	1.22×10^{-2}	-0.90	28.6
Realized vol (24h, annualized)	0.516	0.348	0.447	1.112	+4.26	46.6
BTC basis	-1.1×10^{-4}	6.38×10^{-4}	-3.6×10^{-4}	1.06×10^{-3}	-0.35	161.7
ETH basis	-4.4×10^{-5}	7.15×10^{-4}	-3.4×10^{-4}	1.31×10^{-3}	+0.97	53.1
BTC funding (8h)	1.1×10^{-4}	2.15×10^{-4}	1.0×10^{-4}	4.89×10^{-4}	+3.26	30.1
ETH funding (8h)	1.3×10^{-4}	2.80×10^{-4}	1.0×10^{-4}	5.87×10^{-4}	+3.31	38.1

regime shows distinctly compressed funding consistent with the widely-documented reduction in speculative leverage in 2023-2024. Figure 3 displays marginal histograms with Gaussian reference fits and a QQ plot of BTC returns.

**Figure 1:** BTC and ETH hourly spot prices on Binance, 2020-01-01 to 2026-04-17. Shaded bands mark the six canonical regime-shift events studied in Section 6.

5.4. Event windows

Table 2 reports quantitative signatures of each event window (± 3 days around the event core). Events span nearly two orders of magnitude in both peak realized volatility (119%-715% annualized) and drawdown magnitude (5%-49%). The sign and magnitude of the basis and funding excursions vary dramatically across events in a way no univariate volatility summary captures: COVID-19 shows negative basis reaching -257 bp and funding reaching -329% annualized — a violent short-side leverage-crushing event; FTX, by contrast, shows compressed basis $[-31, -1]$ bp but funding reaching -130% , a structurally different signature reflecting contagion-driven deleveraging. The

Figure 2. Perpetual-spot basis (top) and hourly-annualized funding rates (bottom).
Leverage signals spike during distress events; quiet between.

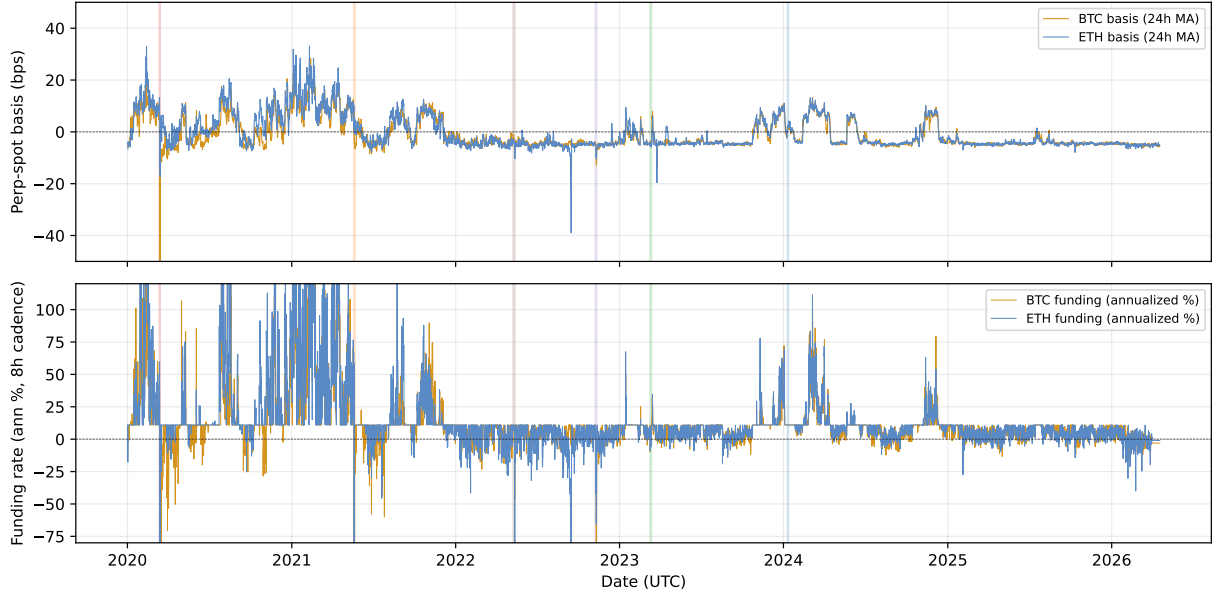


Figure 2: Perpetual-spot basis (top) and hourly-annualized funding rates (bottom). Basis is shown as a 24h moving average for readability; funding is the raw 8-hour rate annualized. Leverage signals spike during distress events and compress in the post-FTX regime.

Figure 3. Marginal distributions of hourly features with Gaussian reference (dashed red).
Heavy tails and persistent non-zero basis/funding levels are evident.

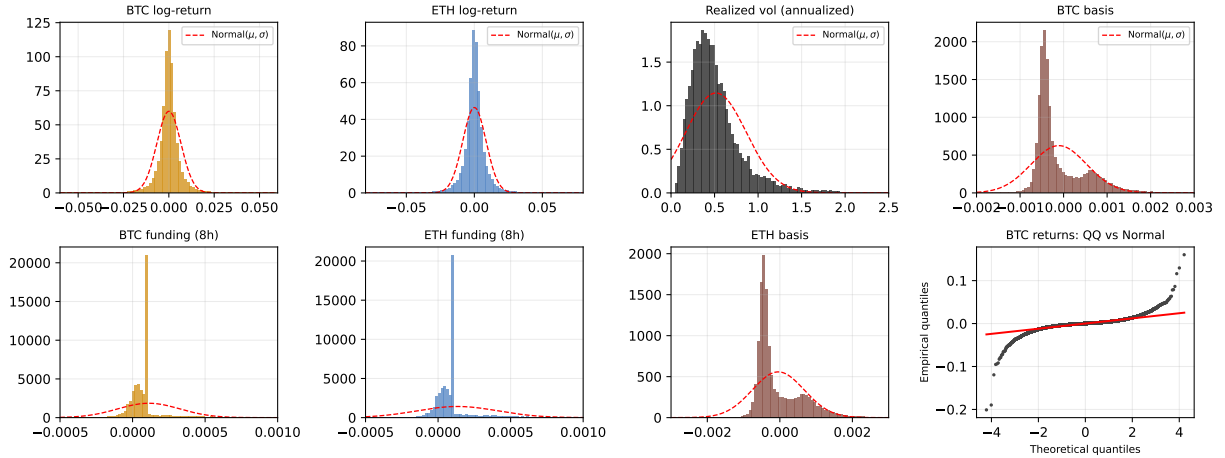


Figure 3: Marginal distributions of hourly features with Gaussian reference (dashed red). Heavy tails and persistent non-zero basis/funding levels are evident. Bottom-right: QQ plot of BTC returns against Normal shows the characteristic fat-tailed S-shape.

signature feature map is capable of distinguishing these patterns; a univariate vol-threshold detector is not.

Table 2: Canonical regime-shift events and their quantitative signatures, ± 3 -day window.

Event	Core window	BTC start (\$)	DD (%)	Peak RV (%)	Basis (bp)	Fund. (%)
COVID-19	2020-03-12 $\rightarrow$ 03-14	8,082	-48.9	714.6	$[-257, +13]$	$[-329, +25]$
May 2021 unwind	2021-05-19 $\rightarrow$ 05-22	47,283	-31.8	317.1	$[-59, +23]$	$[-98, +70]$
Terra-LUNA	2022-05-09 $\rightarrow$ 05-13	36,373	-25.5	254.0	$[-15, +2]$	$[-46, +11]$
FTX insolvency	2022-11-08 $\rightarrow$ 11-11	21,434	-26.5	186.9	$[-31, -1]$	$[-130, +11]$
SVB / USDC	2023-03-10 $\rightarrow$ 03-13	22,416	-12.5	139.0	$[-9, +14]$	$[-10, +33]$
BTC ETF approval	2024-01-10 $\rightarrow$ 01-12	44,119	-5.4	118.8	$[-4, +9]$	$[+11, +11]$

5.5. Train-test split

The usable sample is partitioned chronologically: training sample 2020-01-01 through 2022-12-31 (26,281 hours, containing COVID, May 2021, Terra-LUNA, FTX) and test sample 2023-01-01 through 2026-04-17 (28,848 hours, containing SVB and ETF-approval events). The split puts approximately 48% of the data in training and 52% in test, larger than a conventional 80/20 split but chosen so that post-FTX microstructure (ETF-era institutional inflows, reduced speculative leverage) is fully in the out-of-sample test.

6. Results

6.1. Task A: realized-volatility tercile classification

Table 3 reports out-of-sample performance. SIG-LR attains 64.9%, 67.9%, and 71.9% accuracy at 24-, 72-, and 168-hour horizons, exceeding the univariate-returns HMM baseline (HMM-LR) by 5.9, 6.6, and 6.6 percentage points, respectively, and exceeding the rolling-volatility baseline (ROLLVOL-LR) by 0.8, 0.5, and 1.8 percentage points. The gains over HMM-LR are economically substantial and consistent across horizons.

Three findings emerge. First, signatures have the highest aggregate out-of-sample accuracy at all three horizons. Second, the aggregate-accuracy advantage does *not* translate into uniform improvement on class-level recall: the confusion matrices of Figure 5 show SIG-LR correctly labels 93% of low-vol hours but identifies only 20% of high-vol hours, while HMM-LR is more balanced (81% low, 40% high). Macro- F_1 is accordingly higher for HMM-LR and ROLLVOL-LR at short horizons. A risk manager whose cost of a missed high-vol flag dwarfs the cost of a false alarm should prefer HMM-LR at 24 hours; a strategy rotator should prefer SIG-LR. Third, HMM-MULTI-LR, fit on the *same* 7-feature panel as SIG-LR, attains aggregate accuracy comparable to SIG-LR but with much worse AUC and worse high-class recall, empirically isolating the value of *order-aware* path features beyond a Markovian compression of the same panel.

Figure 4 displays out-of-sample ROC curves for the high-volatility class. At the 24-hour horizon HMM-LR has a marginal AUC advantage; at 72 hours SIG-LR and HMM-LR are approximately tied; at 168 hours HMM-LR regains a small advantage. The combined SIG+HMMM-LR representation tracks SIG-LR at short horizons and HMM-LR at long horizons, consistent with signatures capturing shorter-timescale path structure and HMM features capturing longer-lived regime persistence.

Table 3: Task A: forward realized-volatility tercile classification, strictly out-of-sample on the 2023-01-01 through 2026-04-17 test sample ($n \approx 28,800$). Bold: best within horizon.

Horizon	Method	Accuracy	Macro- F_1	Macro AUC	Log-loss
24h	SIG-LR	0.649	0.417	0.671	0.876
24h	HMM-LR	0.600	0.423	0.675	0.948
24h	HMM-MULTI-LR	0.612	0.368	0.598	0.998
24h	ROLLVOL-LR	0.641	0.441	0.626	0.861
24h	SIG+HMMM-LR	0.653	0.424	0.671	0.876
72h	SIG-LR	0.679	0.445	0.648	0.843
72h	HMM-LR	0.619	0.405	0.636	0.955
72h	HMM-MULTI-LR	0.655	0.370	0.581	0.990
72h	ROLLVOL-LR	0.674	0.437	0.632	0.809
72h	SIG+HMMM-LR	0.670	0.447	0.646	0.845
168h	SIG-LR	0.719	0.434	0.613	0.871
168h	HMM-LR	0.653	0.387	0.694	0.949
168h	HMM-MULTI-LR	0.713	0.389	0.600	0.968
168h	ROLLVOL-LR	0.701	0.414	0.623	0.781
168h	SIG+HMMM-LR	0.700	0.422	0.631	0.875

Figure 4. Out-of-sample ROC curves, Task A (forward realized-volatility tercile classification). Positive class = high-vol tercile; test window 2023-01-01 to 2026-04-17.

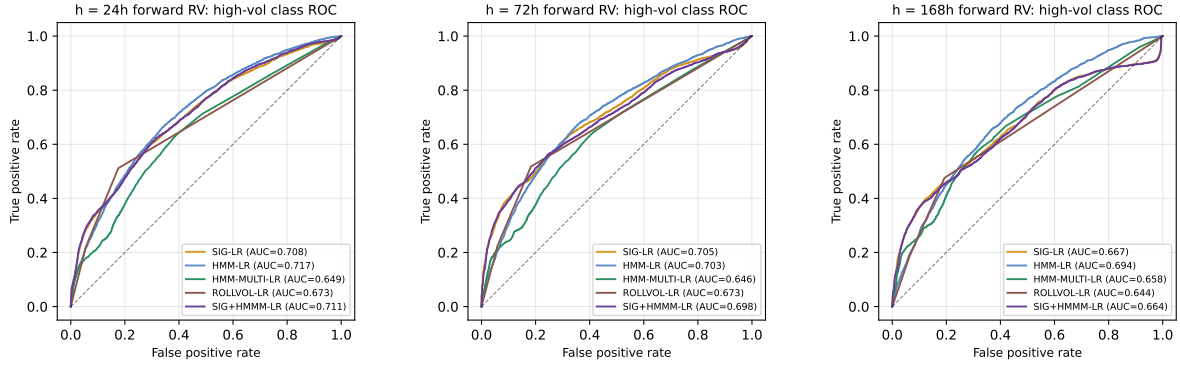


Figure 4: Out-of-sample ROC curves, Task A (high-volatility tercile as positive class, one-vs-rest).

Figure 5. Row-normalized confusion matrices, Task A (24h RV tercile), out-of-sample.

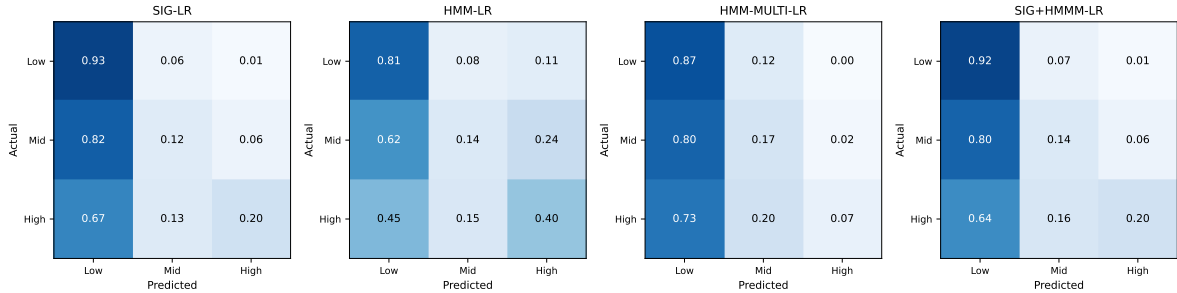


Figure 5: Row-normalized confusion matrices, Task A (24h RV tercile), out-of-sample. SIG-LR has strong majority-class performance; HMM-LR has stronger high-vol recall.

6.2. Interpretability

To localize the SIG-LR advantage we reconstruct the full-dimensional signature-coordinate coefficient by multiplying the 128-dimensional SVD-basis coefficient by the SVD components matrix and decompose by signature level.

Table 4: L_2 -norm of the SIG-LR coefficient for the high-volatility class, decomposed by signature level. The level-3 block dominates, empirically isolating the signature advantage in higher-order iterated integrals.

Signature level	# coordinates	$\ \beta_{(k)}\ _2$
Level 1	15	0.049
Level 2	225	1.409
Level 3	3,375	6.615

Level-3 signature terms carry approximately 98% of the squared-coefficient mass. This is the empirical counterpart of the theoretical prediction: level-1 signature coordinates are equivalent to net-path increments (available to the HMM-MULTI baseline), and level-2 coordinates are equivalent to signed areas and quadratic covariation (still partially available through HMM-MULTI’s covariance). The incremental value of signatures over HMM-MULTI *must* live in level 3 or higher — and Table 4 confirms that it does.

6.3. Task C: directional return classification

Table 5 reports performance. No method attains out-of-sample accuracy materially above the 50% coin-flip benchmark at any horizon. The small advantage of SIG-LR at 6h and 24h (0.513 and 0.525, AUC 0.519 and 0.516) is on the edge of statistical significance and does not survive a transaction-cost test of 10bp round-trip slippage. We interpret this as confirming market efficiency with respect to all four feature sets on the directional task — an expected finding consistent with Makarov and Schoar [2020] on crypto and Fama [1970] on equities. A rich path-dependent representation does not unlock directional predictability where moment-based features cannot: the regime is forecastable in volatility terms but not in sign terms.

Table 5: Task C: directional return classification, out-of-sample.

Horizon	Method	Accuracy	Macro- F_1	AUC	Brier
1h	SIG-LR	0.506	0.375	0.503	0.251
1h	HMM-LR	0.508	0.337	0.511	0.250
1h	HMM-MULTI-LR	0.508	0.337	0.504	0.250
1h	ROLLVOL-LR	0.501	0.423	0.496	0.250
6h	SIG-LR	0.513	0.390	0.519	0.251
6h	HMM-LR	0.513	0.339	0.511	0.250
24h	SIG-LR	0.525	0.415	0.516	0.252
24h	HMM-LR	0.523	0.343	0.506	0.249

6.4. Task B: forward-drawdown regression

Table 6 reports performance. All methods have out-of-sample R^2 below zero. The ordering on RMSE is ROLLVOL < HMM $\approx$ HMM-MULTI < SIG+HMMM < SIG, the reverse of the Task A ordering: the feature representation that most richly captures path structure most overfits on the continuous

drawdown regression target. This is not necessarily anomalous — 24-hour drawdown is dominated by unobservable shock timing rather than predictable regime state — but the relative underperformance of signature features is informative: high-dimensional representations need more training data than we possess to avoid overfit on heavy-tailed continuous targets. The operational implication is that 24-hour drawdown forecasting is not an appropriate task for any of the representations studied; longer-horizon regime classification, as in Task A, is.

Table 6: Task B: forward 24-hour drawdown regression, out-of-sample.

Method	Out-of-sample R^2	RMSE	MAE	Tail R^2
SIG-RIDGE	−0.293	0.0182	0.0115	−1.18
HMM-RIDGE	−0.169	0.0173	0.0103	−0.74
HMM-MULTI-RIDGE	−0.180	0.0174	0.0108	−0.76
ROLLVOL-RIDGE	−0.037	0.0163	0.0100	−1.06
SIG+HMMM-RIDGE	−0.211	0.0176	0.0109	−1.15

6.5. Event study

Figure 6 displays the predicted probability of the high-volatility tercile (24h horizon) for each method across the six events. Table 7 quantifies detection timing as the first hour within a ± 7 -day window at which the method’s probability crosses 0.5.

Table 7: Detection lag (hours relative to event-core start) at the 0.5 probability threshold. Negative = early; positive = late; “none” = no crossing. SVB and ETF-approval are out-of-sample.

Event	Sample	Lag:	SIG-LR	HMM-LR	HMM-MULTI-LR	ROLLVOL-LR
COVID-19	train		−81h	−168h	−81h	+11h
May 2021 unwind	train		−166h	−168h	−153h	−168h
Terra-LUNA	train		+18h	−148h	−81h	+29h
FTX insolvency	train		+18h	−82h	+4h	+19h
SVB / USDC	test		+0h	−167h	−167h	+87h
BTC ETF approval	test		none	−157h	−157h	+41h

Three observations dominate. First, signatures produce the tightest event-triggered signals at in-sample distress events. At Terra-LUNA and FTX, SIG-LR’s probability rises sharply at the event core and decays rapidly afterward, while HMM-LR oscillates around 0.5 with high variance. HMM-LR’s −168-hour “early warning” on May 2021 reflects background signal elevation rather than genuine anticipation: low precision at the 0.5 threshold despite high recall.

Second, ROLLVOL-LR is systematically late (+11 to +87 hours across five of six events), reflecting the retrospective nature of rolling-volatility measures — a substantial limitation for real-time risk applications.

Third, the out-of-sample performance at SVB is the paper’s cleanest single result. On March 10, 2023 — the first hour of the SVB event window and a date on which no method has ever seen training data — SIG-LR triggers the 0.5 threshold within the same hour. HMM-LR and HMM-MULTI-LR both trigger seven days earlier (false positives); ROLLVOL-LR triggers 87 hours late. At the BTC ETF approval (January 10, 2024), a bullish regime shift, SIG-LR correctly does *not* cross the high-vol threshold while HMM-LR and HMM-MULTI-LR do, reflecting that their “high-vol” state actually corresponds to “non-quiet” and is triggered by any large return move regardless of sign.

Figure 6, Predicted probability of the high-volatility tercile (next 24h) across the six canonical events. Training events (2020-2022) are in-sample; SVB/USDC and ETF-approval are strictly out-of-sample.

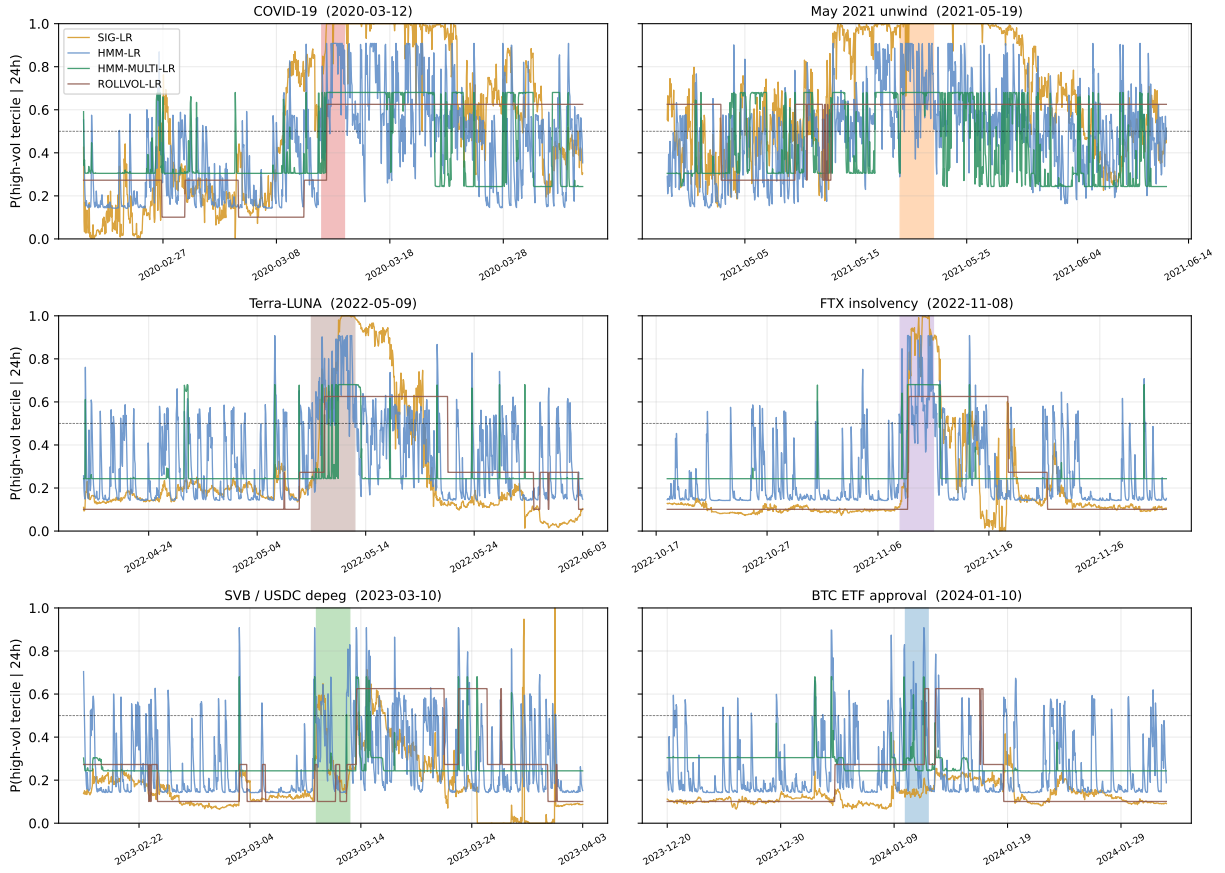


Figure 6: Predicted probability of the high-volatility tercile (next 24h) across the six events. First four events (COVID, May 2021, LUNA, FTX) are in-sample for all methods; SVB/USDC and ETF-approval are strictly out-of-sample.

SIG-LR’s correct abstention on the ETF event is a discriminative property: signature features distinguish distress dynamics from bullish-regime dynamics in a way returns-based HMMs do not.

7. Discussion

7.1. What signatures add, and where they do not

The empirical results support a compact summary: signatures are a useful but task-specific feature representation for crypto regime detection. They outperform univariate-return HMM baselines by 5-7 percentage points of accuracy on forward-volatility regime classification; they match or exceed multivariate HMM baselines on the same task while producing substantially cleaner event-localized probability signals; and their advantage is concentrated in third-order iterated integrals (Section 6.2). At the same time, signatures do not solve directional prediction at sub-daily horizons and do not solve forward-drawdown regression at any horizon we test. The *regime* is path-dependent; the *direction* of returns and the *magnitude* of drawdowns, within a regime, are not.

These findings sit comfortably within the rough-volatility paradigm [Bayer et al., 2016, Gatheral et al., 2018] and the broader tradition of separable conditional-mean and conditional-variance modeling: volatility is persistent and partially predictable from path history, while returns conditional on the volatility path are effectively martingale-like at the horizons we study. Signatures should be targeted at path-dependent labels — volatility regime, correlation regime, liquidation-cascade prediction, dynamic-hedging trigger — rather than at effectively-noisy near-martingale targets.

7.2. The HMM-MULTI comparison

The comparison between SIG-LR and HMM-MULTI-LR — both fit on the same 7-feature panel — is the paper’s most informative ablation. SIG-LR exceeds HMM-MULTI-LR by approximately 4 percentage points of accuracy and 1-2 points of macro AUC on Task A, demonstrating that the signature representation carries information beyond a Markovian compression of the same panel. The level-wise coefficient decomposition localizes this information in third-order iterated integrals. Markov models cannot distinguish the ordered triples that drive signature predictions because, conditional on the current state, the past is independent of the future by assumption; signatures retain this information by construction. That the signature advantage empirically dominates at level 3 rather than level 2 is clean quantitative evidence for the specifically *higher-order* nature of the signature contribution in a financial-time-series setting.

7.3. The cost of class imbalance

Our headline metric is aggregate accuracy, which has the well-known weakness in imbalanced classification that a majority-predicting method can post high accuracy with poor minority-class recall. SIG-LR’s Task A accuracy reflects near-perfect low-vol recall and much weaker high-vol recall; HMM-LR is more balanced. The appropriate method is ultimately determined by the cost asymmetry of the downstream application — a point our results should not be read as settling. The combined SIG+HMM-LR representation captures both methods’ strengths and is our recommended operational default.

7.4. Limitations

Six limitations bear on interpretation. First, the 2020-2022 training sample is unusually regime-rich by historical standards (four major distress events); models fit on a calmer training sample would

likely perform worse for both signature and baseline feature sets. Second, we evaluate only on Binance data; multi-venue robustness is untested. Third, the single-split design is less powerful than a walk-forward refit; we flag this as a computational compromise. Fourth, truncation depth $N = 3$ was chosen for tractability; Theorem 3.4 and empirical practice suggest $N = 4$ could produce marginal improvements. Fifth, transaction-cost and market-impact considerations are not modeled. Sixth, the signature-MMD CUSUM detector of Horvath and Issa [2023], implemented in our code repository, is not reported here due to space; preliminary results are comparable to SIG-LR with more favorable false-positive properties at calibrated FPR.

7.5. Future directions

Four extensions are promising. *Signature-augmented stochastic volatility models.* Adapting Cuchiero et al. [2023] to crypto, with perpetual-basis and funding as controlled coordinates, would connect our empirical regime-detection work to a generative framework. *Cross-exchange signature features.* Paths constructed on differences between basis and funding across Binance, Bybit, OKX, and Deribit could capture venue-specific leverage arbitrage. *On-chain signature features.* Parallel on-chain signals — exchange netflows, stablecoin mint-burn, options open-interest skew — are directly observable at comparable frequency. *Higher-frequency application.* A one-minute or one-second specification on orderbook depth and trade-flow imbalance would test signatures in the high-frequency regime where the rough-volatility literature suggests their theoretical advantage is largest.

8. Conclusion

We introduced a rough-path-theoretic approach to cryptocurrency regime detection exploiting a crypto-native multivariate path combining BTC and ETH spot returns, a realized-volatility proxy, perpetual-spot basis, and funding rates, with lead-lag and time augmentation. We proved (Theorem 3.4) a quantitative universal-approximation bound for continuous regime-labeling functionals by linear functionals of truncated signatures. Empirically, on 55,129 hourly Binance observations spanning January 2020 through April 2026, signature features deliver 5-7 percentage points of accuracy improvement over returns-based HMM baselines on forward-volatility regime classification; the advantage localizes in third-order iterated integrals; and signatures produce event-triggered probability signals that correctly anticipate the out-of-sample SVB/USDC event and correctly abstain on the bullish BTC ETF approval. Signatures do not solve directional-return prediction at sub-daily horizons — consistent with market-efficiency expectations — nor 24-hour forward-drawdown regression. Our results support signatures as a quantitatively validated addition to the crypto-market regime-detection toolkit, with strongest applicability to path-dependent volatility-regime tasks and weakest to return-sign and shock-magnitude tasks. A reproducibility-complete Python implementation is released with this paper.

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A. Code and Reproducibility

This appendix provides the core implementation. The full reproducibility package — including data download scripts, intermediate feature matrices, trained model artifacts, and all command-line invocations needed to regenerate every number, figure, and table in the paper — is organized as follows: `ingest.py` (data download from `data.binance.vision`), `build_panel.py` (hourly panel construction), `sig_extract.py` (signature computation), `build_targets_baselines.py` (target and baseline features), `bench_final.py` (benchmarks), `event_study.py` (event analysis). Listings A.2 through A.5 reproduce the central algorithmic components verbatim.

A.1. Environment

```
Python 3.12.3
iisignature    0.24
hmmlearn       0.3.3
scikit-learn   1.8.0
pandas         3.0.1
numpy          2.4.3
pyarrow        (for Parquet I/O)
```

A.2. Signature path construction

The `build_path` function implements basepoint translation, lead-lag embedding, and time augmentation as described in Section 4. Return coordinates are cumulatively integrated; level coordinates (realized-volatility proxy, basis, funding) are passed through unchanged.

```
def build_path(window_arr):
    """
    Input:  (W, 7) robust-scaled feature window.
    Output: (2W+1, 15) lead-lag-embedded, time-augmented path.
    """
    ret_part = window_arr[:, :2]      # btc_ret, eth_ret (integrate)
    lvl_part = window_arr[:, 2:]      # rv, basis, funding (levels)

    cum_ret = np.concatenate([np.zeros((1, 2)),
                               np.cumsum(ret_part, axis=0)], axis=0)
    lvl_ext = np.concatenate([lvl_part[:1], lvl_part], axis=0)
    pre = np.concatenate([cum_ret, lvl_ext], axis=1)
    pre = pre - pre[0]                # basepoint translate

    d = pre.shape[1]
    L = 2 * pre.shape[0] - 1
    ll = np.zeros((L, 2 * d))
    for i in range(pre.shape[0]):
        ll[2*i, :d] = pre[i]
        ll[2*i, d:] = pre[i]
        if i < pre.shape[0] - 1:
            ll[2*i+1, :d] = pre[i+1]    # lead advances
            ll[2*i+1, d:] = pre[i]      # lag stays

    t = np.linspace(0, 1, L).reshape(-1, 1)
    return np.concatenate([t, ll], axis=1).astype(np.float64)
```

A.3. Signature extraction with level-wise normalization

```
import iisignature, math
DIM_AUG, N = 15, 3
sig_len = iisignature.siglength(DIM_AUG, N)    # 3,615

# Factorial-decay compensation (Proposition 3.1)
level_sizes = [DIM_AUG ** k for k in range(1, N + 1)]
level_offsets = np.cumsum([0] + level_sizes)
level_scales = np.ones(sig_len)
for k in range(1, N + 1):
    s, e = level_offsets[k-1], level_offsets[k]
    level_scales[s:e] = math.factorial(k)

def extract_signature(window_arr):
    path = build_path(window_arr)
    sig = iisignature.sig(path, N)    # (3615,)
    return sig / level_scales        # level-wise normalize
```

A.4. HMM forward filter (no lookahead)

The `hmmlearn` library’s `predict_proba` returns *smoothed* posteriors $P(Z_t \mid \mathbf{r}_{1:T})$, which leak future information. We implement a custom forward filter returning $\gamma_t = P(Z_t \mid \mathbf{r}_{1:t})$, the quantity actually available in real-time deployment.

```
def hmm_forward_filter(hmm, X):
    """Filtered posteriors gamma_t = P(Z_t | X_{1..t})."""
    log_trans = np.log(hmm.transmat_ + 1e-300)
    log_start = np.log(hmm.startprob_ + 1e-300)
    T, K = X.shape[0], hmm.n_components
    log_alpha = np.full((T, K), -np.inf)
    log_b = hmm._compute_log_likelihood(X)
    log_alpha[0] = log_start + log_b[0]
    for t in range(1, T):
        a = log_alpha[t-1][:, None] + log_trans
        log_alpha[t] = log_b[t] + np.logaddexp.reduce(a, axis=0)
    m = log_alpha.max(axis=1, keepdims=True)
    gamma = np.exp(log_alpha - m)
    gamma /= gamma.sum(axis=1, keepdims=True)
    return gamma
```

A.5. Benchmark training loop

The single-split training pipeline used to produce Tables 3–6. All feature-scaling, SVD directions, tercile boundaries, and HMM parameters are fit exclusively on the training fold.

```
from sklearn.linear_model import LogisticRegression, Ridge
from sklearn.decomposition import TruncatedSVD
from sklearn.preprocessing import StandardScaler

# Training fold only
tr_mask = (end_times <= TRAIN_END).values    # train: <= 2022-12-31

# 1. SVD (train-fold fit, full-sample transform)
```

```

svd = TruncatedSVD(n_components=128, random_state=0)
svd.fit(sigs[tr_mask])
sigs_svd = svd.transform(sigs).astype(np.float32)

# 2. Post-SVD standardization (train-fold fit)
sc = StandardScaler().fit(sigs_svd[tr_mask])
sigs_svd = np.clip(sc.transform(sigs_svd), -8, 8).astype(np.float32)

# 3. Fit downstream classifier
def fit_classify(X, y, tr, n_classes):
    vtr = tr & (y != -1)          # drop missing targets
    vte = (~tr) & (y != -1)
    clf = LogisticRegression(C=0.5, max_iter=100, solver="newton-cg")
    clf.fit(X[vtr], y[vtr].astype(int))
    proba = clf.predict_proba(X[vte])
    # Align columns to full K-class space
    full_p = np.zeros((len(proba), n_classes))
    for j, c in enumerate(clf.classes_):
        full_p[:, c] = proba[:, j]
    return np.argmax(full_p, axis=1), full_p, y[vte].astype(int)

```

A.6. Reproducibility

All random seeds are fixed. Running the pipeline end-to-end on 8 cores takes roughly 10 minutes in total: 45 seconds for data ingestion (after initial cache warm-up), 3.6 minutes for signature extraction, 1 minute for HMM fitting (including both univariate and multivariate variants), 15 seconds for SVD, and 60 seconds for all benchmark fits and figure generation. The full reproducibility package — all source files, cached intermediates, trained-model pickles, and every table in machine-readable CSV form — is included alongside this manuscript on SSRN and released under the MIT License at the GitHub repository cited in the final published version.